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(54) **DELIVERY SYSTEM FOR VALVE REPAIR**

(56) **References Cited**

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**FOREIGN PATENT DOCUMENTS**

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|    |            |    |         |
|----|------------|----|---------|
| CN | 109498216  | A  | 3/2019  |
| CN | 110313947  | A  | 10/2019 |
| CN | 111358509  | A  | 7/2020  |
| CN | 111568605  | A  | 8/2020  |
| CN | 212234802  | U  | 12/2020 |
| CN | 115153958  | A  | 10/2022 |
| CN | 115245407  | A  | 10/2022 |
| CN | 115281893  | A  | 11/2022 |
| CN | 117462303  | A  | 1/2024  |
| WO | 2022143141 | A1 | 7/2022  |
| WO | 2024139057 | A1 | 7/2024  |

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**OTHER PUBLICATIONS**

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(57) **ABSTRACT**

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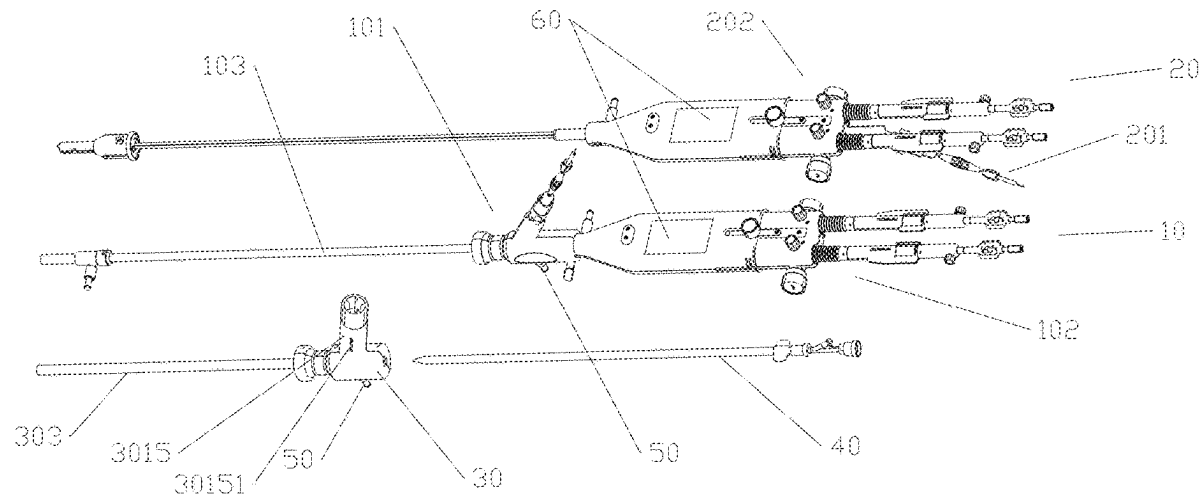
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**A61F 2/24** (2006.01)

A delivery system for valve repair is provided, and relates to the technical field of valve repair. The first delivery assembly includes a first wire locker control device and a first nailing control device, one end of the first wire locker control device is fixedly connected with the multi-lumen tube, and the other end is detachably connected with the first nailing control device; the second delivery assembly is detachably connected with the first wire locker control device. The second delivery assembly includes a second wire locker control device and a second nailing control device. The multi-lumen tube internally includes a first lumen tube, a second lumen tube and a third lumen tube which are axially arranged.

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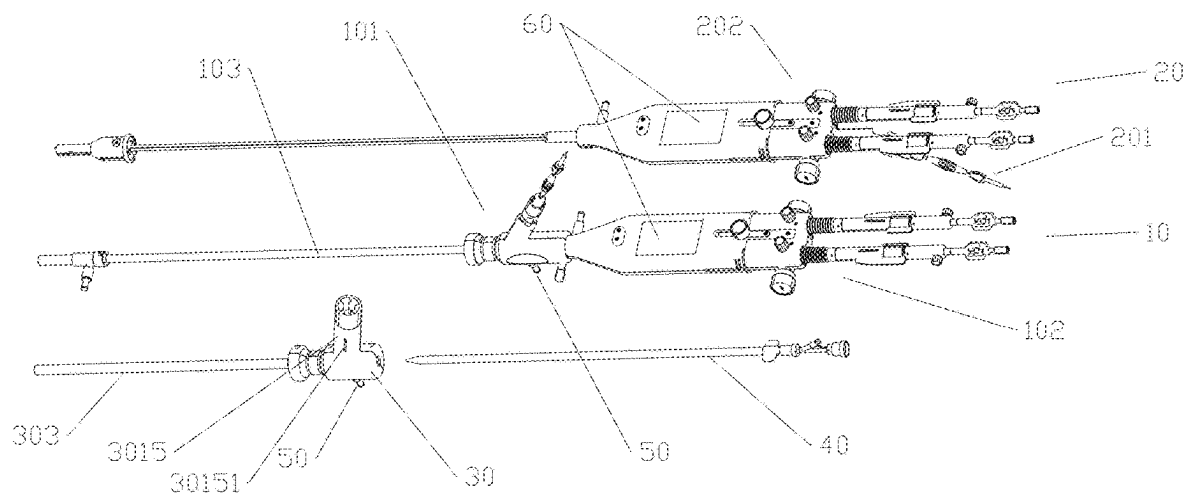


Fig. 1

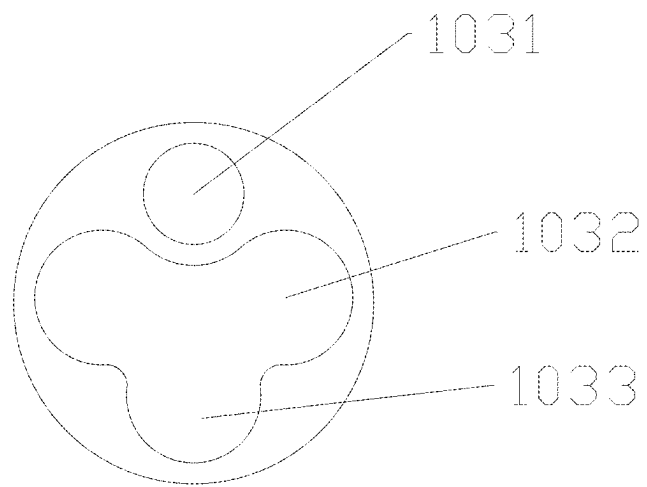


Fig. 2

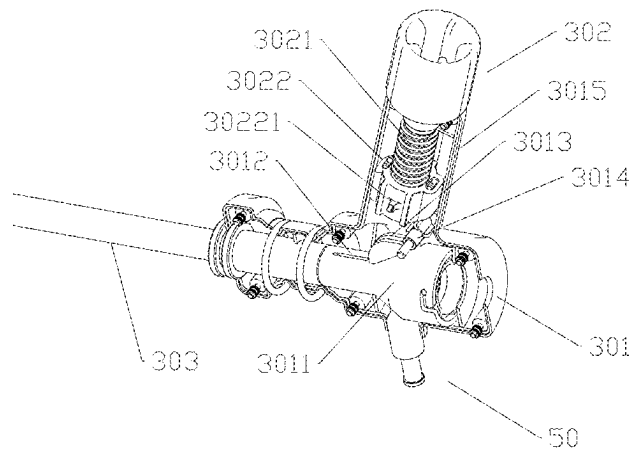


Fig. 3

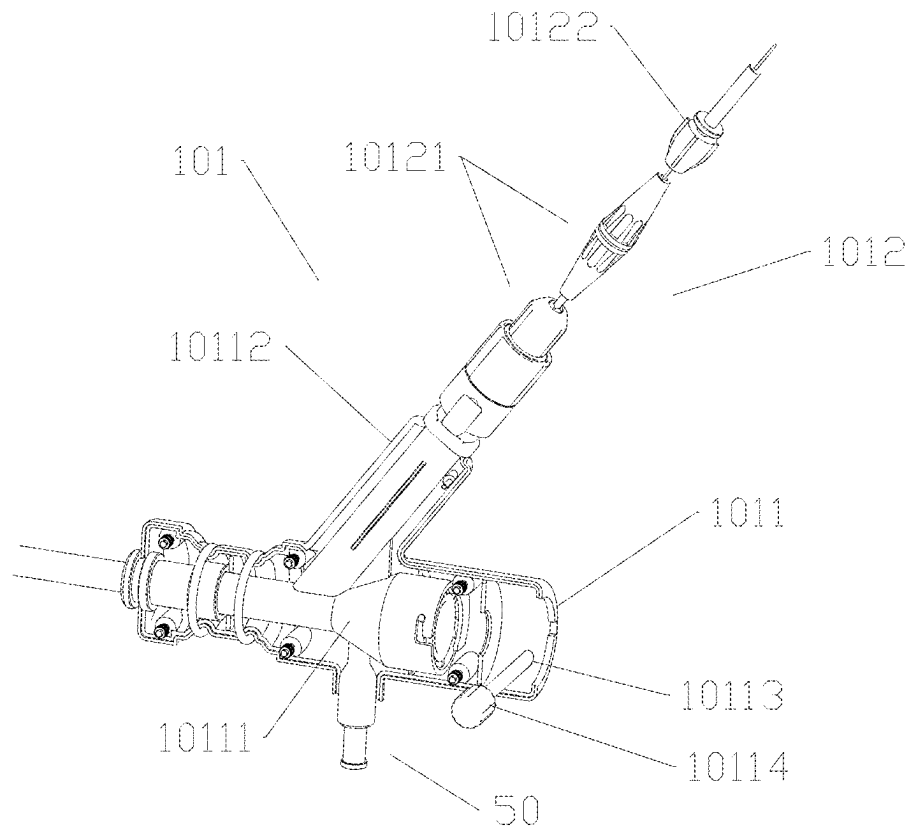


Fig. 4

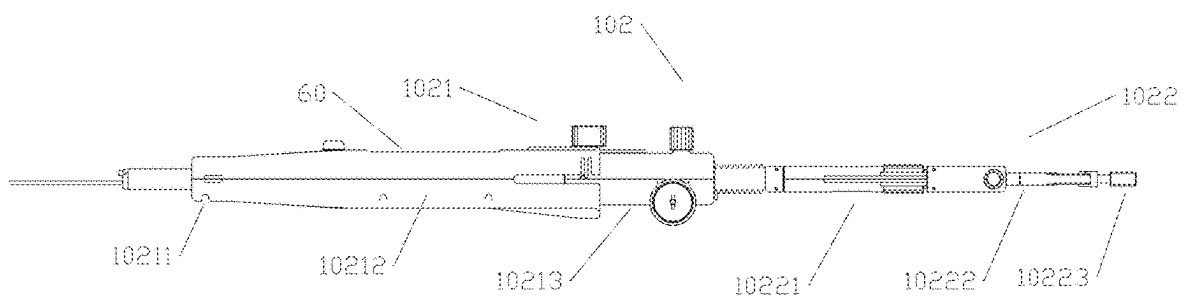


Fig. 5

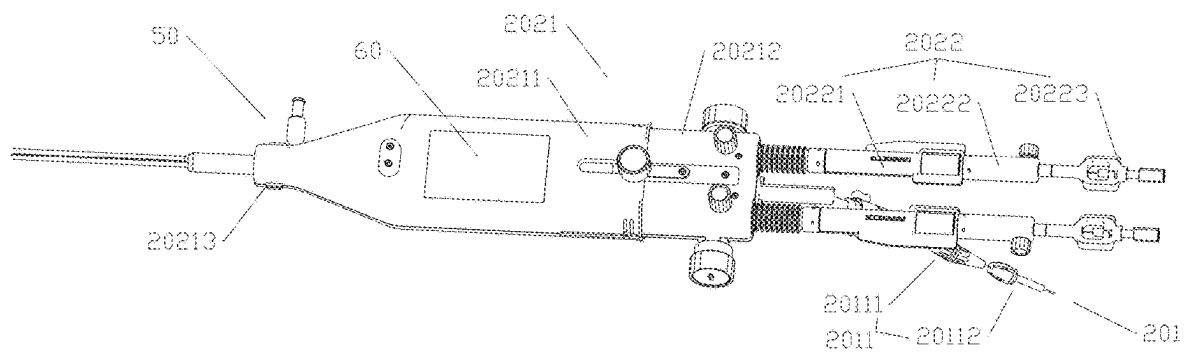


Fig. 6

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**DELIVERY SYSTEM FOR VALVE REPAIR****CROSS-REFERENCE TO RELATED APPLICATION**

This application is based on and claims priority under 35 U.S.C. 119 to Chinese Patent Application No. 202411495978.X, filed on Oct. 25, 2024, in the China National Intellectual Property Administration. The entire disclosure of the above application is incorporated herein by reference.

**TECHNICAL FIELD**

The disclosure relates to the technical field of valve repair, in particular to a delivery system for valve repair.

**BACKGROUND**

Heart valves are barriers between different cardiovascular structures. They can only open and close in a certain direction to ensure that blood can only flow forward and not backward. The mitral valve is a “one-way valve” between the left atrium and the left ventricle, ensuring that blood circulation flows from the left atrium to the left ventricle in a directional manner and passes a certain amount of blood flow. Blood flows from the left atrium through the mitral valve into the left ventricle, and is then pumped into the aorta by the left ventricle and flows to the whole body. The mitral valve is opened, and blood flows from the left atrium into the left ventricle. Then the mitral valve is closed to ensure that when the left ventricle contracts to pump blood to the aorta, blood does not flow back into the left atrium. If the mitral valve is diseased and cannot be fully closed, it will cause blood to flow back into the left atrium when the left ventricle contracts. This is mitral regurgitation.

The mitral regurgitation is the most common heart valve disease worldwide. Regurgitation reduces blood flow to various parts of the body. In order to compensate, the heart will try to pump blood more forcefully, which increases the workload of the heart. Patients with severe mitral regurgitation may experience a variety of debilitating symptoms, such as shortness of breath, palpitations, dizziness and fatigue. These patients face the risk of poor quality of life, significant limitation of activities, repeated hospitalizations due to heart failure and increased mortality. Chronic severe mitral regurgitation is often accompanied by heart failure, which can lead to death if not treated.

**SUMMARY**

The delivery system for valve repair of the present disclosure includes:

a first delivery assembly and a second delivery assembly; the first delivery assembly includes a first wire locker control device and a first nailing control device, one end of the first wire locker control device is fixedly connected to a multi-lumen tube, and the other end of the first wire locker control device is detachably connected to the first nailing control device, the first nailing control device is configured to control a first nailing device to nail, and the first wire locker control device is configured to control a first wire locker to lock the first nailing device;

the second delivery assembly is detachably connected to the first wire locker control device, the second delivery assembly includes a second wire locker control device

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and a second nailing control device, the second nailing control device is configured to control a second nailing device to nail, and the second wire locker control device is configured to control a second wire locker to lock the second nailing device;

the second wire locker control device is arranged on a housing of the second nailing control device, and the second nailing control device is detachably connected to the first wire locker control device; based on that the second nailing control device is connected to the first wire locker control device, the first nailing control device is detached from the first wire locker control device;

the multi-lumen tube includes a first lumen, a second lumen and a third lumen which are arranged axially, the first lumen is used for the first wire locker to enter, the second lumen is used for the first nailing device and the second nailing device to enter respectively, and the third lumen is used for the second wire locker to enter; the delivery system for valve repair is configured to enable the first wire locker and the second wire locker to synchronously lock the first nailing device and the second nailing device through the first lumen and the third lumen.

The delivery system for valve repair of the present disclosure further includes a sheath tube bending-adjustment assembly;

the sheath tube bending-adjustment assembly includes a first mounting bracket and a first adjustment structure, wherein the first adjustment structure is arranged on a side of the first mounting bracket for adjusting a bending of a sheath tube, one end of the first mounting bracket is connected to the sheath tube, and the other end of the first mounting bracket is connected to the first wire locker control device;

the multi-lumen tube passes through the first mounting bracket and passes through the sheath tube.

In the delivery system for valve repair, the first wire locker control device includes a second mounting bracket and a second adjustment structure, one end of the second mounting bracket is connected to the multi-lumen tube, and the other end of the second mounting bracket is connected to the first nailing control device or the second delivery assembly, the second adjustment structure is fixedly connected to a side of the second mounting bracket, and the second adjustment structure is configured to control an axial movement and a circumferential rotation of the first wire locker;

the first wire lock passes through the second adjustment structure and the second mounting bracket in sequence.

In the delivery system for valve repair, the first nailing control device includes a third mounting bracket and a third adjustment structure, one end of the third mounting bracket is connected to the second mounting bracket, and the other end of the third mounting bracket is connected to the third adjustment structure;

the first nailing device passes through the third adjustment structure, the third mounting bracket and the second mounting bracket in sequence, and the third adjustment structure is configured to control an axial movement and a circumferential rotation of the first nailing device.

In the delivery system for valve repair, the second nailing control device includes a fourth mounting bracket and a fourth adjustment structure, one end of the fourth mounting bracket is connected to the second mounting bracket, and the



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other end of the fourth mounting bracket is connected to the fourth adjustment structure and the second wire locker control device;

the second nailing device passes through the fourth adjustment structure, the fourth mounting bracket and the second mounting bracket in sequence, and the fourth adjustment structure is configured to control an axial movement and a circumferential rotation of the second nailing device.

In the delivery system for valve repair, the second wire locker control device includes a fifth adjustment structure, and the fifth adjustment structure is connected to the fourth mounting bracket;

the second wire locker passes through the fifth adjustment structure, the fourth mounting bracket and the second mounting bracket in sequence, and the fifth adjustment structure is configured to control an axial movement and a circumferential rotation of the second wire locker.

In the delivery system for valve repair, a first limiting hole is arranged on a side wall of the second mounting bracket, the first limiting hole penetrates the side wall of the second mounting bracket, a first limiting groove is arranged on a side wall of the third mounting bracket, and a second limiting groove is arranged on a side wall of the fourth mounting bracket;

based on that the third mounting bracket or the fourth mounting bracket is connected to the second mounting bracket, a first limiting member is inserted into the first limiting groove or the second limiting groove through the first limiting hole, and the first limiting member is configured to limit a movement of the third mounting bracket or a movement of the fourth mounting bracket.

The delivery system of valve repair further includes a dilation tube, wherein the dilation tube is detachably connected to the sheath tube bending-adjustment assembly.

In the delivery system for valve repair, injection channels are respectively arranged on the second mounting bracket, the third mounting bracket and the fourth mounting bracket, and the injection channels are configured to inject sealing liquid to seal an internal space of the multi-lumen tube.

In the delivery system for valve repair according to claim 5, observation windows are respectively arranged on the third mounting bracket and the fourth mounting bracket, and the observation windows are made of transparent material.

#### BRIEF DESCRIPTION OF FIGURES

Other features, objects and advantages of the present disclosure will become more apparent by reading the detailed description of non-limiting embodiments made with reference to the following drawings:

FIG. 1 schematically shows a schematic diagram of a structure of a delivery system according to an embodiment of the present disclosure;

FIG. 2 schematically shows a cross-sectional view of a multi-lumen tube according to an embodiment of the present disclosure;

FIG. 3 schematically shows a schematic diagram of a structure of a sheath tube bending-adjustment assembly according to an embodiment of the present disclosure;

FIG. 4 schematically shows a schematic diagram of a structure of a first wire lock control device according to an embodiment of the present disclosure;

FIG. 5 schematically shows a schematic diagram of a structure of a first nailing control device according to an embodiment of the present disclosure;

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FIG. 6 schematically shows a schematic diagram of a structure of a second delivery assembly according to an embodiment of the present disclosure;

#### DETAILED DESCRIPTION

In order to make the purpose, technical solution and advantages of the embodiments of the present disclosure clearer, the technical solution in the embodiments of the present disclosure will be clearly and completely described below in conjunction with the drawings in the embodiments of the present disclosure. Obviously, the described embodiments are part of the embodiments of the present disclosure, not all of the embodiments. Based on the embodiments of the present disclosure, all other embodiments obtained by ordinary technicians in this field without creative work are within the scope of protection of the present disclosure.

The terms used in the embodiments of the present disclosure are only for the purpose of describing specific embodiments, and are not intended to limit the present disclosure. The singular forms “a”, “said” and “the” used in the embodiments of the present disclosure and the appended claims are also intended to include plural forms, unless the context clearly indicates other meanings.

It should be noted that the directional words such as “upper”, “lower”, “left”, and “right” described in the embodiments of the present disclosure are described at the angles shown in the drawings and should not be understood as limiting the embodiments of the present disclosure. In addition, in the context, it should also be understood that when it is mentioned that an element is formed “on” or “under” another element, it can not only be formed directly “on” or “under” another element, but also be formed “on” or “under” another element indirectly through an intermediate element.

At present, there are two main treatment methods for mitral regurgitation: medication and surgery. The surgical treatment is divided into edge-to-edge repair, valve repair and chord repair. For valve repair, multiple groups of anchors are inserted into the valve ring, and each group of anchors is locked by multiple lockers to achieve the overall repair of the valve. However, the internal situation of the heart valve is not visible during treatment, and the locking force is controlled by the doctor's experience. However, locking each group of anchors one by one cannot ensure that the locking force is consistent each time, resulting in an irregular shape of the repaired valve ring, which leads to uneven valve closure along the valve commissure. Tighter closure will reduce blood flow and cause insufficient blood supply, while looser closure will not achieve the purpose of treating reflux.

FIG. 1 schematically shows a schematic diagram of the structure of a delivery system according to an embodiment of the present disclosure, and FIG. 2 schematically shows a cross-sectional view of a multi-lumen tube according to an embodiment of the present disclosure. As shown in FIG. 1 and FIG. 2, a delivery system for valve repair according to the present disclosure includes a first delivery assembly 10 and a second delivery assembly 20.

The first delivery assembly 10 includes a first wire locker control device 101 and a first nailing control device 102. One end of the first wire locker control device 101 is fixedly connected to the multi-lumen tube 103, and the other end is detachably connected to the first nailing control device 102. The first nailing control device 102 is configured to control the first nail driving device to drive nails, and the first wire

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locker control device **101** is configured to control the first wire locker to lock the first nail driving device.

The second delivery assembly **20** is detachably connected to the first wire locker control device **101**. The second delivery assembly **20** includes a second wire locker control device **201** and a second nailing control device **202**. The second nailing control device **202** is configured to control the second nail driving device to drive nails. The second wire locker control device **201** is configured to control the second wire locker to lock the second nail driving device.

The multi-lumen tube **103** includes a first lumen **1031**, a second lumen **1032** and a third lumen **1033** arranged axially therein. The first lumen **1031** is used for the first wire locker to enter, the second lumen **1032** is used for the first nail driver and the second nail driver to enter respectively, and the third lumen **1033** is used for the second wire locker to enter.

The delivery system is configured to allow the first wire locker and the second wire locker to synchronously lock the first nail driver and the second nail driver through the first lumen **1031** and the third lumen **1033**.

Specifically, the first delivery assembly **10** includes a first wire locker control device **101** and a first nailing control device **102**. One end of the first wire locker control device **101** is detachably connected to the first nailing control device **102** and the second delivery assembly **20**, and the other end is fixedly connected to the multi-lumen tube **103**.

When the delivery system is used, the first nailing control device **102** is connected to the first wire locker control device **101**, so that the first wire locker penetrates into the first lumen **1031** of the multi-lumen tube **103**, and the first nailing device penetrates into the second lumen **1032**. When the first nailing device completes the nailing operation, the first nailing control device **102** and the first wire locker control device **101** are disassembled, and the second delivery assembly **20** is installed with the first wire locker control device **101**, so that the second wire locker in the second wire locker control device **201** penetrates into the third lumen **1033**, and the second nailing device in the second nailing control device **202** penetrates into the second lumen **1032**. At this time, the first wire locker penetrates into the first lumen **1031**, the second nailing device penetrates into the second lumen **1032**, and the second wire locker penetrates into the third lumen **1033**.

When the second nailing device completes the nailing operation, the first wire locker control device **101** and the second wire locker control device **201** are synchronously controlled to respectively control the first wire locker to lock the first nailing device and the second wire locker to lock the second nailing device, thereby synchronously completing two sets of locking operations.

With such arrangement, when the doctor synchronously controls the two sets of wire lockers for locking, the locking degrees of the two sets of wire lockers are adjusted in real time by comparing the control degrees of the two sets of wire lockers, so that the locking forces of the two sets of nailing devices are consistent or similar, thereby making the overall shape of the heart valve regular, solving the problem that the doctor cannot refer to each other when locking separately and cannot ensure the consistency of the locking forces twice, resulting in irregular shape of the repaired heart valve and affecting the subsequent recovery degree.

Further, FIG. 3 schematically shows a schematic diagram of the structure of a sheath tube bending-adjustment assembly according to an embodiment of the present disclosure. As shown in FIG. 1 and FIG. 3, the delivery system for valve

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repair according to the present disclosure further includes a sheath tube bending-adjustment assembly **30** and a dilation tube **40**.

The sheath tube bending assembly **30** includes a first mounting bracket **301** and a first adjustment structure **302**. The first adjustment structure **302** is arranged on the side of the first mounting bracket **301** to be used for bending the sheath tube **303**. One end of the first mounting bracket **301** is connected to the sheath tube **303**, and the other end is connected to the first wire locker control device **101**.

The multi-lumen tube **103** passes through the first mounting bracket **301** and through the sheath tube **303**.

Specifically, a first auxiliary connecting piece **3011** is provided in the first mounting bracket **301**. The sheath tube **303** is inserted into the first mounting bracket **301** and fixedly connected to one end of the first auxiliary connecting piece **3011**, and the other end of the first auxiliary connecting piece **3011** is used for inserting the dilation tube **40** and the first wire locker control device **101**, so that the whole is assembled.

A first through hole **3012** and a first groove **3013** are provided on the side wall of the first auxiliary connecting piece **3011**. A guide column **3014** is provided on the side wall of the first mounting bracket **301**. The first through hole **3012** penetrates the side wall of the first auxiliary connecting piece **3011**, and the bending guide wire on the sheath tube **303** can be led out of the first auxiliary connecting piece **3011** through the first through hole **3012**.

The first groove **3013** extends in the axial direction. The first groove **3013** and the first through hole **3012** are on the same axis. After the bending guide wire in the sheath tube **303** is led out through the first through hole **3012**, it is inserted from one end of the first groove **3013** and passes through the other end of the first groove **3013**.

The guide column **3014** is arranged near the other end of the first groove **3013**, and the guide column **3014** extends in the radial direction. When the bending adjustment guide wire passes through the other end of the first groove **3013**, it is wrapped around the side wall of the guide column **3014**. The side wall of the guide column **3014** guides the bending adjustment guide wire of the sheath tube **303** so that the bending adjustment guide wire of the sheath tube **303** changes from extending in the axial direction to extending in the radial direction, and then is fixedly connected to the first adjustment structure **302** located on the side of the first mounting bracket **301**. The first adjustment structure **302** can make the bending adjustment guide wire move back and forth in the radial direction, and then the bending of the sheath tube **303** is achieved by pulling the bending adjustment guide wire.

The first mounting bracket **301** further includes a first housing **3015** located at a side of the first mounting bracket **301** and extending outwards. The first adjustment structure **302** is inserted into the first housing **3015** and fixedly connected to the bending adjustment guide wire.

The first adjustment structure **302** includes a first rotating member **3021** and a second rotating member **3022**. The second rotating member **3022** is sleeved on the first rotating member **3021**, and the two are threadedly connected. A first protrusion **30221** is provided on the circumferential side wall of the second rotating member **3022**. A second groove **30151** extending along the axial direction of the first housing **3015** is provided on the first housing **3015**. The second groove **30151** penetrates the side wall of the first housing **3015**. The first protrusion **30221** is inserted into the second groove **30151** and is slidably connected thereto.

When the first rotating member **3021** is controlled to rotate, the second rotating member **3022** sleeved thereon is limited by the mutual positioning between the first protrusion **30221** and the second groove **30151** to achieve axial movement. The bending adjustment guide wire of the sheath **303** is fixedly connected to the second rotating member **3022**. When the second rotating member **3022** moves, the bending adjustment guide wire will be pulled synchronously, thereby realizing the bending operation of the sheath tube **303**.

In this way, by setting the first adjustment structure **302** for bending sheath **303** on the side of the first mounting bracket **301**, the overall length of the sheath tube bending-adjustment assembly **30** can be shortened, and the length of the wire locker spring tube and the nailing bending-adjustment tube can be reduced to a certain extent. The shorter the length of the wire locker spring tube and the nailing bending-adjustment tube is, the higher the control accuracy of the end control of the wire locker spring tube and the nailing bending-adjustment tube is, which can achieve more precise control during surgery.

Furthermore, an injection channel **50** is also provided on the side of the first auxiliary connecting member **3011**, and the injection channel **50** penetrates through the first mounting bracket **301**. The injection channel **50** is configured to inject sealing liquid to seal the internal space of the first auxiliary connecting member **3011** to prevent external air from entering the human body through the sheath tube **303**, or blood in the human body from flowing out through the sheath tube **303**.

Furthermore, the dilation tube **40** is detachably connected to the sheath tube bending-adjustment assembly **30**, and the dilation tube **40** can be inserted into the sheath tube **303** and partially extend out of the sheath tube **303**.

The insertion end of the dilation tube **40** is set to be conical, which is easier to penetrate the body tissue, guide the insertion of the sheath tube **303**, and provide an insertion path for the sheath tube **303**. After the dilation tube completes the corresponding operation, it is disassembled from the sheath tube bending-adjustment assembly **30**, and the first delivery assembly **10** and the sheath tube bending-adjustment assembly **30** are installed to complete the subsequent valve repair operation.

Further, FIG. 4 schematically shows a schematic diagram of a structure of a first wire locker control device according to an embodiment of the present disclosure, as shown in FIG. 4:

the first wire locker control device **101** includes a second mounting bracket **1011** and a second adjustment structure **1012**. One end of the second mounting bracket **1011** is connected to the multi-lumen tube **103**, and the other end is connected to the first nailing control device **102** or the second delivery assembly **20**. The second adjustment structure **1012** is fixedly connected to the side of the second mounting bracket **1011**. The second adjustment structure **1012** is configured to control the axial movement and circumferential rotation of the first wire locker.

The first wire locker passes through the second adjustment structure **1012** and the second mounting bracket **1011** in sequence.

Specifically, a second auxiliary connecting piece **10111** is provided in the second mounting bracket **1011**. The multi-lumen tube **103** is inserted into the second mounting bracket **1011** and fixedly connected to one end of the second auxiliary connecting piece **10111**. The first nailing control

device **102** and the second delivery assembly **20** are detachably connected to the other end of the second auxiliary connecting piece **10111**.

The second mounting bracket **1011** further includes a second housing **10112** disposed at a side thereof. The second housing **10112** extends in a radial direction. The second adjustment structure **1012** is mounted on the second housing **10112**. A first wire locker spring tube is disposed in the second adjustment structure **1012**. The first wire locker spring tube penetrates the second adjustment structure **1012**, passes through the second housing **10112** and enters into the first lumen **1031** of the multi-lumen tube **103**.

A first wire locker is arranged in the first wire locker spring tube, and the axial movement and circumferential rotation of the first wire locker spring tube can be achieved through the control of the second adjustment structure **1012**, thereby achieving the locking operation of the first wire locker.

A first steel cable is arranged in the first wire locker spring tube, and an end portion of the first steel cable is detachably connected to the first wire locker. The second adjustment structure **1012** includes a first control member **10121** and a second control member **10122**. The first control member **10121** is sleeved on the first wire locker spring tube, and the second control member **10122** is sleeved on the first steel cable. The first control member **10121** can control the axial movement and circumferential rotation of the first wire locker spring tube. The second control member **10122** can control the axial movement and circumferential rotation of the first steel cable, and can control the disassembly of the first steel cable and the first wire locker. When the first wire locker completes the locking operation, the first steel cable and the first wire locker are controlled to be disassembled, and then the first wire locker spring tube and the first steel cable are controlled to be synchronously withdrawn from the body to complete the surgical operation.

An injection channel **50** is also provided on the side of the second auxiliary connecting piece **10111**. The injection channel **50** penetrates through the second mounting bracket **1011**. The injection channel **50** is configured to inject sealing liquid to seal the internal space of the second auxiliary connecting piece **10111** to prevent external air from entering the human body through the multi-lumen tube **103**, or blood from flowing out of the human body.

In this way, by arranging the second adjustment structure **1012** on the side of the first mounting bracket **301**, the length of the entire device can also be shortened, thereby reducing the overall length of the wire locker spring tube and the overall length of the nailing bending-adjustment tube to achieve more precise control.

Further, FIG. 5 schematically shows a schematic diagram of a structure of a first nailing control device according to an embodiment of the present disclosure, as shown in FIG. 5:

the first nailing control device **102** includes a third mounting bracket **1021** and a third adjustment structure **1022**. One end of the third mounting bracket **1021** is connected to the second mounting bracket **1011**, and the other end is connected to the third adjustment structure **1022**.

The first nailing device passes through the third adjustment structure **1022**, the third mounting bracket **1021** and the second mounting bracket **1011** in sequence. The third adjustment structure **1022** is configured to control the axial movement and circumferential rotation of the first nailing device.

Specifically, a first limiting hole **10113** is provided on the side wall of the second mounting bracket **1011**, and a first

limiting groove **10211** is provided on the side wall of the third mounting bracket **1021**. When the first nailing control device **102** and the first wire locker control device **101** are installed, the first limiting groove **10211** and the first limiting hole **10113** are provided correspondingly. The first limiting member **10114** is inserted into the first limiting groove **10211** through the first limiting hole **10113**, so as to limit the movement of the third mounting bracket **1021** relative to the second mounting bracket **1011**, thereby realizing the fixed connection between the first wire locker control device **101** and the first nailing control device **102**.

Furthermore, a first nailing bending-adjustment tube is provided in the first nailing control device **102**, and a first nailing device is provided in the first nailing bending-adjustment tube. When the first nailing control device **102** and the first wire locker control device are installed, the first nailing bending-adjustment tube extends into the second lumen **1032** of the multi-lumen tube **103**, and the first nailing control device **102** controls the axial movement, circumferential rotation and end bending of the first nailing bending-adjustment tube, so that the internal first nailing device is nailed into the valve tissue to realize the nailing operation.

The third mounting bracket **1021** includes a third housing **10212** and a fourth housing **10213**. The third housing **10212** is slidably connected to the fourth housing **10213**, the third housing **10212** is connected to the second mounting bracket **1011**. The first limiting groove **10211** is arranged on the side wall of the third shell **10212**.

The third adjustment structure **1022** is connected to the fourth housing **10213**. When the fourth housing **10213** slides relative to the third housing **10212**, the third adjustment structure **1022** installed on the fourth housing **10213** can move synchronously as a whole.

A first nailing spring-tube is arranged in the first nailing bending-adjustment tube. A second steel cable is arranged in the first nailing spring-tube. The end of the second steel cable is detachably connected to the first nailing device. The third adjustment structure **1022** includes a third control member **10221**, a fourth control member **10222** and a fifth control member **10223**. The third control member **10221** is sleeved with the first nailing bending-adjustment tube, and can control the axial movement, circumferential rotation and bending of the end of the first nailing bending-adjustment tube. The fourth control member **10222** is sleeved with the first nailing spring-tube, and can control the axial movement and circumferential rotation of the first nailing spring-tube. The fifth control member **10223** is sleeved with the second steel cable, and can control the axial movement and circumferential rotation of the second steel cable.

When performing the nailing operation, the third control member **10221** and the fourth control member **10222** are used to control the first nailing bending-adjustment tube and the first nailing spring-tube to move to the nailing position, and the fifth control member **10223** is used to control the second steel cable to rotate circumferentially, so that the first nailing device at the end is nailed into the valve tissue. After completion, the second steel cable is controlled to be disassembled from the first nailing device, and the first nailing bending-adjustment tube and the first nailing spring-tube are controlled to be withdrawn from the body for subsequent operations.

Further, FIG. 6 schematically shows a schematic diagram of a structure of a second delivery assembly according to an embodiment of the present disclosure, as shown in FIG. 6: the second delivery assembly **20** includes a second wire locker control device **201** and a second nailing control

device **202**. The second wire locker control device **201** is arranged on the housing of the second nailing control device **202**.

When the second delivery assembly **20** is installed with the first wire locker device **101**, the second nailing control device **202** is connected with the first wire locker device **101**.

After the first nailing operation of the first nailing device is completed, the first nailing control device **102** and the first wire locker control device **101** are disassembled, and the first nailing bending-adjustment tube is removed from the second lumen **1032** of the multi-lumen tube **103**. At this time, the first nailing device is in the human body, and the first wire locker spring-tube is still in the first lumen **1031** of the multi-lumen tube **103**, and the first nailing device is not locked.

At this time, the second nailing control device **202** is installed with the first wire locker control device **101**, the second nailing bending-adjustment tube in the second nailing control device **202** is extended into the second lumen **1032**, and the second wire locker spring-tube in the second wire locker control device **201** is extended into the third lumen **1033** of the multi-lumen tube **103**. When the second nailing control device **202** controls the second nailing device to complete the nailing operation, the first wire locker control device **101** and the second wire locker control device **201** are synchronously controlled to respectively make the first wire locker lock the first nailing device and the second wire locker lock the second nailing device.

In this way, by arranging the first lumen **1031**, the second lumen **1032**, and the third lumen **1033** in the multi-lumen tube **103**, when the first wire locker control device **101** and the second wire locker control device **201** are synchronously operated, the specific locking force of each wire locker can be clearly perceived by comparing the two controls, and whether the locking force of a certain wire locker is larger or smaller can be perceived, and real-time adjustments can be achieved, thereby making the final shape of the valve regular after repair.

Furthermore, the second nailing control device **202** includes a fourth mounting bracket **2021** and a fourth adjustment structure **2022**. One end of the fourth mounting bracket **2021** is connected to the second mounting bracket **1011**, and the other end is connected to the fourth adjustment structure **2022** and the second wire locker control device **201**.

The second nailing device passes through the fourth adjustment structure **2022**, the fourth mounting bracket **2021** and the second mounting bracket **1011** in sequence. The fourth adjustment structure **2022** is configured to control the axial movement and circumferential rotation of the second nailing device.

Specifically, the fourth mounting bracket **2021** has the same structure as the third mounting bracket **1021**, and is also provided with a fifth housing **20211** and a sixth housing **20212**. The sixth housing **20212** is slidably connected to the fifth housing **20211**. The fourth adjustment structure **2022** and the second wire locker control device **201** are installed on the sixth housing **20212**. When the sixth housing **20212** moves relative to the fifth housing **20211**, the fourth adjustment structure **2022** and the second wire locker control device **201** can be controlled to move synchronously.

A second limiting groove **20213** is provided on the side wall of the fifth housing **20211**. When the fifth housing **20211** and the second mounting bracket **1011** are installed, the first limiting member **10114** is inserted into the first limiting hole **10113** and the second limiting groove **20213** to

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achieve fixed connection between the fourth mounting bracket **2021** and the second mounting bracket **1011**.

The fourth adjustment structure **2022** has the same structure as the third adjustment structure **1022**. The second nailing bending-adjustment tube passes through the fourth adjustment structure **2022** and the fourth mounting bracket **2021** and enters the second lumen tube **1032** of the multi-lumen tube **103**. A second nailing spring-tube is arranged in the second nailing bending-adjustment tube. A third steel cable is arranged in the second nailing spring-tube. The end of the third steel cable is detachably connected to a second nailing device.

The fourth adjustment structure **2022** includes a sixth control member **20221**, a seventh control member **20222** and an eighth control member **20223**. The sixth control member **20221** is sleeved on the second nailing bending-adjustment tube, which can control the axial movement, circumferential rotation and end bending of the second nailing bending-adjustment tube. The seventh control member **20222** is sleeved on the second nailing spring-tube, which can control the axial movement and circumferential rotation of the second nailing spring-tube. The eighth control member **20223** is sleeved on the third steel cable, which can control the axial movement and circumferential rotation of the third steel cable to realize the nailing operation of the second nailing device.

Furthermore, the second wire lock control device **201** includes a fifth adjustment structure **2011**, and the fifth adjustment structure **2011** has the same structure as the second adjustment structure **1012**.

A second wire locker spring-tube is arranged in the second wire locker control device **201**. The second wire locker spring-tube passes through the fifth adjustment structure **2011**, and passes through the fourth mounting bracket **2021** and the second mounting bracket **1011** and extends into the third lumen **1033** of the multi-lumen tube **103**.

A fourth steel cable is arranged in the second wire locker spring-tube. A second wire locker is arranged at the end of the fourth steel cable. The fifth adjustment structure **2011** includes a ninth control member **20111** and a tenth control member **20112**. The ninth control member **20111** is sleeved on the second wire locker spring-tube, and can control the axial movement and circumferential rotation of the second wire locker spring-tube. The tenth control member **20112** is sleeved on the fourth steel cable, and can control the axial movement and circumferential rotation of the fourth steel cable, so as to realize the operation of the second wire locker locking the second nailing device.

Furthermore, an observation window **60** is provided on the third mounting bracket **1021** and the fourth mounting bracket **2021**. The observation window **60** is made of a transparent material. The first nailing bending-adjustment tube in the first nailing control device **102** can pass through the observation window **60**, and the second nailing bending-adjustment tube and the second wire locker spring-tube in the second nailing control device **202** can pass through the observation window **60**, so that the doctor can check the axial movement distance of the tube.

Furthermore, the present disclosure also provides a control method of a delivery system, which is applied to the above-mentioned delivery system for valve repair, and the method includes:

S01: inserting the dilation tube **40** into the sheath tube bending-adjustment assembly **30**, and guiding the sheath tube **303** into the body;

S02: disassembling the dilation tube **40**, installing the first wire locker control device **101** and the sheath tube

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bending-adjustment assembly **30** to insert the multi-lumen tube **103** into the sheath tube **303**, and insert the first wire locker spring-tube into the first lumen **1031** of the multi-lumen tube **103**;

S03: installing the first nailing control device **102** and the first wire locker control device **101**, so that the first nailing bending-adjustment tube is inserted into the second lumen tube **1032** of the multi-lumen tube **103**;

S04: controlling the first nailing control device **102** to nail the first nailing device in the first nailing bending-adjustment tube into the valve tissue, disassembling the first nailing control device **102** and the first wire locker control device **101**, and withdrawing the first nailing bending-adjustment tube from the second lumen **1032**;

S05: installing the second nailing control device **202** and the first wire locker control device **101**, insert the second nail driving bending adjustment tube into the second lumen **1032**, and insert the second wire locker spring tube into the third lumen **1033**;

S06: controlling the second nailing control device **202** to drive the second nailing device in the second nailing bending-adjustment tube into the valve tissue;

S07: synchronously controlling the first wire locker control device **101** and the second wire locker control device **201** to respectively make the first wire locker in the first wire locker spring-tube lock the first nailing device and the second wire locker in the second wire locker spring-tube lock the second nailing device with the same or similar locking force;

S08: withdrawing the sheath tube bending-adjustment assembly **30** from the body, and disassembling the connections among the sheath tube bending-adjustment assembly **30**, the first wire locker **101**, and the second nailing control device **202** to complete the surgical operation.

With such arrangement, through the above control method, the synchronous locking of the two sets of nailing devices can be achieved, which solves the problem that when the locking of the nailing devices is controlled individually, there is no comparison between the two sets of nailing devices, and the locking force of each set cannot be referenced to each other for real-time adjustment, resulting in irregular shape of the repaired valve. Through the scheme of the present disclosure, the sheath adjustment assembly includes a first mounting bracket and a first adjustment structure, and the first adjustment structure is arranged on the side of the first mounting bracket. In this way, the overall length of the sheath adjustment device can be reduced, thereby shortening the length of the nailing and bending adjustment tube and the wire locking spring tube. The shorter the length of the nailing and bending adjustment tube and the wire locking spring tube, the higher the accuracy of the first delivery assembly and the second delivery assembly in controlling their movement and bending.

Moreover, after step S07, if a third set of nailing is required on the valve, the first wire locking control device **101** and the sheath tube bending-adjustment assembly **30** can be disassembled, and the third delivery assembly having a third wire locker control device and a third nailing control device and the sheath tube bending-adjustment assembly **30** can be installed to complete the third set of nailing operation.

To sum up, through the scheme of the present disclosure, when the first nailing device completes nailing, the first wire locker can remain in the first lumen **1031** of the multi-lumen tube **103**. When the second nailing device nails, the second wire locker is located in the third lumen **1033**. When the

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second nailing device completes nailing, the doctor can synchronously control the first wire locker control device **101** and the second wire locker control device **201** to control the first wire locker and the second wire locker to respectively lock the first nailing device and the second nailing device with the same or similar locking force, so that the shape of the repaired valve is regular.

The above description is only a preferred embodiment of the present disclosure and an explanation of the technical principles used. Those skilled in the art should understand that the scope of the disclosure involved in the present disclosure is not limited to the technical solution formed by a specific combination of the above technical features, but should also cover other technical solutions formed by any combination of the above technical features or their equivalent features without departing from the inventive concept. For example, the above features are replaced with (but not limited to) technical features with similar functions disclosed in the present disclosure.

What is claimed is:

**1.** A delivery system for valve repair, comprising:  
a first delivery assembly and a second delivery assembly;  
wherein the first delivery assembly comprises a first wire locker control device and a first nailing control device, one end of the first wire locker control device is fixedly connected to a multi-lumen tube, and the other end of the first wire locker control device is detachably connected to the first nailing control device, the first nailing control device is configured to control a first nailing device to nail, and the first wire locker control device is configured to control a first wire locker to lock the first nailing device;  
the second delivery assembly is detachably connected to the first wire locker control device, the second delivery assembly comprises a second wire locker control device and a second nailing control device, the second nailing control device is configured to control a second nailing device to nail, and the second wire locker control device is configured to control a second wire locker to lock the second nailing device;  
the second wire locker control device is arranged on a housing of the second nailing control device, and the second nailing control device is detachably connected to the first wire locker control device; based on that the second nailing control device is connected to the first wire locker control device, the first nailing control device is detached from the first wire locker control device;  
the multi-lumen tube comprises a first lumen, a second lumen and a third lumen which are arranged axially, the first lumen is used for the first wire locker to enter, the second lumen is used for the first nailing device and the second nailing device to enter respectively, and the third lumen is used for the second wire locker to enter;  
wherein the delivery system for valve repair is configured to enable the first wire locker and the second wire locker to synchronously lock the first nailing device and the second nailing device through the first lumen and the third lumen.

**2.** The delivery system for valve repair according to claim **1**, further comprising a sheath tube bending-adjustment assembly;

wherein the sheath tube bending-adjustment assembly comprises a first mounting bracket and a first adjustment structure, wherein the first adjustment structure is arranged on a side of the first mounting bracket for adjusting a bending of a sheath tube, one end of the first

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mounting bracket is connected to the sheath tube, and the other end of the first mounting bracket is connected to the first wire locker control device;

the multi-lumen tube passes through the first mounting bracket and passes through the sheath tube.

**3.** The delivery system for valve repair according to claim **1**, wherein the first wire locker control device comprises a second mounting bracket and a second adjustment structure, one end of the second mounting bracket is connected to the multi-lumen tube, and the other end of the second mounting bracket is connected to the first nailing control device or the second delivery assembly, the second adjustment structure is fixedly connected to a side of the second mounting bracket, and the second adjustment structure is configured to control an axial movement and a circumferential rotation of the first wire locker;

wherein the first wire lock passes through the second adjustment structure and the second mounting bracket in sequence.

**4.** The delivery system for valve repair according to claim **3**, wherein the first nailing control device comprises a third mounting bracket and a third adjustment structure, one end of the third mounting bracket is connected to the second mounting bracket, and the other end of the third mounting bracket is connected to the third adjustment structure;

wherein the first nailing device passes through the third adjustment structure, the third mounting bracket and the second mounting bracket in sequence, and the third adjustment structure is configured to control an axial movement and a circumferential rotation of the first nailing device.

**5.** The delivery system for valve repair according to claim **4**, wherein the second nailing control device comprises a fourth mounting bracket and a fourth adjustment structure, one end of the fourth mounting bracket is connected to the second mounting bracket, and the other end of the fourth mounting bracket is connected to the fourth adjustment structure and the second wire locker control device;

wherein the second nailing device passes through the fourth adjustment structure, the fourth mounting bracket and the second mounting bracket in sequence, and the fourth adjustment structure is configured to control an axial movement and a circumferential rotation of the second nailing device.

**6.** The delivery system for valve repair according to claim **5**, wherein the second wire locker control device comprises a fifth adjustment structure, and the fifth adjustment structure is connected to the fourth mounting bracket;

wherein the second wire locker passes through the fifth adjustment structure, the fourth mounting bracket and the second mounting bracket in sequence, and the fifth adjustment structure is configured to control an axial movement and a circumferential rotation of the second wire locker.

**7.** The delivery system for valve repair according to claim **5**, wherein a first limiting hole is arranged on a side wall of the second mounting bracket, the first limiting hole penetrates the side wall of the second mounting bracket, a first limiting groove is arranged on a side wall of the third mounting bracket, and a second limiting groove is arranged on a side wall of the fourth mounting bracket;

based on that the third mounting bracket or the fourth mounting bracket is connected to the second mounting bracket, a first limiting member is inserted into the first limiting groove or the second limiting groove through the first limiting hole, and the first limiting member is

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configured to limit a movement of the third mounting bracket or a movement of the fourth mounting bracket.

8. The delivery system for valve repair according to claim 2, further comprising a dilation tube, wherein the dilation tube is detachably connected to the sheath tube bending- 5 adjustment assembly.

9. The delivery system for valve repair according to claim 5, wherein injection channels are respectively arranged on the second mounting bracket, the third mounting bracket and the fourth mounting bracket, and the injection channels are 10 configured to inject sealing liquid to seal an internal space of the multi-lumen tube.

10. The delivery system for valve repair according to claim 5, wherein observation windows are respectively arranged on the third mounting bracket and the fourth 15 mounting bracket, and the observation windows are made of transparent material.

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